predictions of the genes would be wrong, and they would pay the penalty. The young bear would die, and they inside it.

Prediction in a complex world is a chancy business. Every decision that a survival machine takes is a gamble, and it is the business of genes to program brains in advance so that on average they take decisions that pay off. The currency used in the casino of evolution is survival, strictly gene survival, but for many purposes individual survival is a reasonable approximation. If you go down to the water-hole to drink, you increase your risk of being eaten by predators who make their living lurking for prey by water-holes. If you do not go down to the water-hole you will eventually die of thirst. There are risks whichever way you turn, and you must take the decision that maximizes the long-term survival chances of your genes. Perhaps the best policy is to postpone drinking until you are very thirsty, then go and have one good long drink to last you a long time. That way you reduce the number of separate visits to the water-hole, but you have to spend a long time with your head down when you finally do drink. Alternatively the best gamble might be to drink little and often, snatching quick gulps of water while running past the water-hole. Which is the best gambling strategy depends on all sorts of complex things, not least the hunting habit of the predators, which itself is evolved to be maximally efficient from their point of view. Some form of weighing up of the odds has to be done. But of course we do not have to think of the animals as making the calculations consciously. All we have to believe is that those individuals whose genes build brains in such a way that they tend to gamble correctly are as a direct result more likely to survive, and therefore to propagate those same genes.

We can carry the metaphor of gambling a little further. A gambler must think of three main quantities, stake, odds, and prize. If the prize is very large, a gambler is prepared to risk a big stake. A gambler who risks his all on a single throw stands to gain a great deal. He also stands to lose a great deal, but on average high-stake gamblers are no better and no worse off than other players who play for low winnings with low stakes. An analogous comparison is that between speculative and safe investors on the stock market. In some ways the stock market is a better analogy than a casino, because casinos are deliberately rigged in the bank's favour (which means, strictly, that high-stake players will on average end up poorer than low-stake players; and low-stake players poorer than those who do not gamble at all. But this is for a reason not germane to